MATH3611/MATH5705: Problem Sheet 1 Hints

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Q1:

We define a bijection $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$, meaning every natural number z corresponds uniquely to a pair (m, n).

Diagonal Enumeration Strategy

We list all pairs $(m, n) \in \mathbb{N} \times \mathbb{N}$ by sweeping diagonally through a 2D grid. Each diagonal corresponds to a fixed value of $s = m + n$.

- Diagonal $s = 0$: $(0, 0)$
- Diagonal $s = 1$: $(1, 0), (0, 1)$
- Diagonal $s = 2$: $(2, 0), (1, 1), (0, 2)$
- Diagonal $s = 3$: $(3, 0), (2, 1), (1, 2), (0, 3)$
- etc.

Each diagonal s contains $s + 1$ elements. The total number of pairs before diagonal s is the triangular number:

$$T_s = \sum_{i=0}^{s-1}(i + 1) = \frac{s(s + 1)}{2}$$

From $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$

The Cantor pairing function gives an explicit bijection:

$$\pi(m, n) = \frac{(m + n)(m + n + 1)}{2} + n$$

This assigns each pair (m, n) a unique natural number z.

Q4:

Assume $A \neq \emptyset$. Then

There exists an injection $f : A \to B \iff$ There exists a surjection $g : B \to A$.

Proof:

($\Rightarrow$) Suppose there exists an injection $f : A \to B$.

Since f is injective, each element of A corresponds to a distinct element in $f(A) \subseteq B$. Fix an element $a_0 \in A$. Define a function $g : B \to A$ by

$$g(b) = \begin{cases} f^{-1}(b), & \text{if } b \in f(A), \\ a_0, & \text{if } b \notin f(A). \end{cases}$$

For every $a \in A$, there exists $b = f(a) \in f(A)$ such that $g(b) = f^{-1}(f(a)) = a$, so g is surjective onto A.

($\Leftarrow$) Suppose there exists a surjection $g : B \to A$.

For each $a \in A$, since g is surjective, there exists at least one $b_a \in B$ such that $g(b_a) = a$. Define $f : A \to B$ by choosing such an element b_a for each a: $f(a) = b_a$. This f is injective because if $f(a_1) = f(a_2) = b$, then applying g gives $a_1 = g(b) = a_2$. Hence, f is an injection from A into B.

Q8i:

Since the series $\sum x_n$ converges and all $x_n > 0$, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\sum_{n>N} x_n < \varepsilon$. In particular, for $n > N$, each $x_n < \varepsilon$; otherwise, if $x_n \geq \varepsilon$ for some $n > N$, then $\sum_{n>N} x_n \geq x_n \geq \varepsilon$, which contradicts the tail bound. Hence, all $x_n > \varepsilon$ must occur for $n \leq N$. So $A_\varepsilon \subseteq \{1, 2, \ldots, N\}$, and thus A_ε is finite.

Q8ii:

Let $f : [0,1] \to (0, \infty)$. Define $E_n = \{x \in [0,1] \mid f(x) \geq \frac{1}{n}\}$ for $n \in \mathbb{N}$. Since $f(x) > 0$, we have $[0,1] = \bigcup_{n=1}^{\infty} E_n$. As $[0,1]$ is uncountable, if all E_n were countable, their countable union would be countable, a contradiction. Thus, some E_n is uncountable. Set $\epsilon = \frac{1}{n}$ for such an n. Then $\{x \in [0,1] \mid f(x) \geq \epsilon\} = E_n$ is uncountable.

Q9:

Let $S = \{s_1, s_2, s_3, \dots\} \subseteq [0,1]$ be a countable subset, enumerated without repetition. Choose a sequence of positive numbers $(a_n)_{n=1}^{\infty}$ such that

$$a_n > 0, \quad \text{and} \quad \sum_{n=1}^{\infty} a_n = 1.$$

Define the function $f : [0,1] \to [0,1]$ by

$$f(x) = \sum_{\substack{n=1 \\ s_n \le x}}^{\infty} a_n.$$

Properties of f (double-check)

- f is *increasing.*
- f is bounded between 0 and 1.
- At each point $s_n \in S$, f has a jump discontinuity of size a_n:

  $$\lim_{x \to s_n^-} f(x) = \sum_{\substack{k=1 \\ s_k < s_n}}^{\infty} a_k, \quad \lim_{x \to s_n^+} f(x) = \sum_{\substack{k=1 \\ s_k \le s_n}}^{\infty} a_k,$$

  so the jump is $f(s_n^+) - f(s_n^-) = a_n$.
- f is continuous at every point $x \notin S$.